



The House of Wisdom in Baghdad was a major center of Islamic scholarship, attracting the foremost thinkers from across the Islamic world.

THE ARABIC AND PERSIAN EMPIRES had a long tradition of scholarship, and this continued after the birth of the Islamic religion. Islam encouraged scientific and philosophical pursuits, which were not seen as incompatible with theology. **Libraries and other centers of learning** were established in many Islamic cities during the **Islamic “Golden Age.”** Perhaps the greatest of these was the **House of Wisdom** (Bayt al-Hikma), founded in Baghdad at the beginning of the

astrolabe—an instrument used to observe the position of stars. Although not the first to produce a work on the astrolabe, al-Khwarizmi’s contribution was significant, especially in the Islamic world, where the astrolabe could be used to calculate the time of daily prayers.

The Chinese were pioneers in the technology of **printing**, largely due to their invention of paper—possibly as early as the 2nd century BCE—which lent itself better to printing than the papyrus and parchment used

400 THOUSAND THE NUMBER OF BOOKS IN THE HOUSE OF WISDOM

9th century. As well as housing thousands of books, the House of Wisdom **encouraged research** and the **translation** of mathematical, scientific, and philosophical texts from ancient Greece.

Persian mathematician and astronomer Muhammad ibn Musa **al-Khwarizmi** (c.780–850) was one of the most important scholars at the House of Wisdom, studying both Greek and Indian scientific treatises. In around 820, he **described the use of the**

elsewhere. Developing a form of **woodblock printing on silk** that had appeared around 200, they applied the technique to paper and used it for the mass production of books. By the 9th century, it was being used to **print promissory notes** that were in effect a form of paper money issued by the Chinese government.

“ **FONDNESS FOR SCIENCE... HAS ENCOURAGED ME TO COMPOSE A SHORT WORK ON... WHAT IS EASIEST AND MOST USEFUL IN ARITHMETIC.** ”

Al-Khwarizmi, Persian mathematician, c.780–850

ONE OF AL-KHWARIZMI’S MAJOR ACHIEVEMENTS was a treatise on mathematics entitled *The Compendious Book on Calculation by Completion and Balancing*, published around 830. It contained a description of the branch of mathematics now known as **algebra**. Although he drew on sources such as Greek and Indian texts (see 250–500), he is considered to be the inventor of algebra. In his book, al-Khwarizmi explained the process of **balancing both sides of an equation** (*al-jabr* in Arabic, hence the modern term algebra), and gave a systematic way of **solving quadratic equations**, which had been described almost 500 years earlier by Greek mathematician Diophantus of Alexandria. Central to his method was the principle

$$ax^2 + bx + c = 0$$

ALGEBRA

Algebra is a branch of mathematics that uses letters to represent unknown quantities (called variables), and symbols for operations such as addition and subtraction. These can be combined in an algebraic statement known as an expression, such as “ $a + 3$ ”. A mathematical statement, such as “ $a + 3 = 7$ ”, is known as an equation. Equations in which the highest power of an unknown quantity is two are known as quadratic equations (as above), and those in which the highest power is three are called cubic equations.

of balancing an equation by transposing terms from one side to the other and canceling out terms that appear on both sides.

Another prominent scholar at the House of Wisdom was the polymath **Abu Yusuf Ya’qub ibn Ishaq al-Kindi** (also known

as al-Kindi), who in the mid-9th century wrote a large number of **treatises on various scientific subjects**, ranging from mathematics, astronomy, and optics, to medicine and geography. A scholar of theology and philosophy, he was also responsible for the **translation of many classic Greek texts** and their incorporation into Islamic thinking. It is largely through al-Kindi’s translations and commentaries on Indian texts that **Indian numerals were introduced to the Islamic world**, and subsequently became the **basis for the modern system of numbers**, although zero was probably “discovered” later (see 861–99).

Al-Kindi was very **sceptical about alchemy**, refuting one of its central ideas—the

AL-KINDI (c.801–873)

Born and educated in Kufa, near Baghdad, Al-Kindi was one of the first major scholars of the newly founded House of Wisdom. He translated Greek scientific and philosophical texts into Arabic, and incorporated Hellenistic ideas into Islamic scholarship. He wrote treatises on many subjects, including medicine, chemistry, astronomy, and mathematics.



c.810 The Bayt al-Hikma (House of Wisdom) is founded in Baghdad

812 The Chinese government issues a form of paper money

c.820 Al-Khwarizmi describes the astrolabe

c.830 Al-Khwarizmi describes algebra in his *Compendious Book on Calculation by Completion and Balancing*

c.850 Al-Kindi writes treatises on optics, perspective, medicine, and cryptography

855 Chinese alchemists describe the discovery of gunpowder